

A life cycle assessment framework combining nutritional and environmental health impacts of diet: a case study on milk

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Received: 30 January 2015 / Accepted: 2 September 2015 / Published online: 23 September 2015
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Abstract

Purpose While there has been considerable effort to understand the environmental impact of a food or diet, nutritional effects are not usually included in food-related life cycle assessment (LCA).

Methods We developed a novel Combined Nutritional and Environmental Life Cycle Assessment (CONE-LCA) framework that evaluates and compares in parallel the environmental and nutritional effects of foods or diets. We applied this framework to assess human health impacts, expressed in Disability Adjusted Life Years (DALYs), in a proof-of-concept case study that investigated the environmental and nutritional human health effects associated with the addition of one serving of fluid milk to the present average adult US diet. Epidemiology-based nutritional impacts and benefits linked to milk intake, such as colorectal cancer, stroke, and

prostate cancer, were compared to selected environmental impacts traditionally considered in LCA (global warming and particulate matter) carried to a human health endpoint.

Results and discussion Considering potential human health effects related to global warming, particulate matter, and nutrition, within the context of this study, findings suggest that adding one serving of milk to the current average diet could result in a health benefit for American adults, assuming that existing foods associated with substantial health benefits are not substituted, such as fruits and vegetables. The net health benefit is further increased when considering an iso-caloric substitution of less healthy foods (sugar-sweetened beverages). Further studies are needed to test whether this conclusion holds within a more comprehensive assessment of environmental and nutritional health impacts.

Conclusions This case study provides the first quantitative epidemiology-based estimate of the complements and trade-offs between nutrition and environment human health burden expressed in DALYs, pioneering the infancy of a new approach in LCA. We recommend further testing of this CONE-LCA approach for other food items and diets, especially when making recommendations about sustainable diets and food choices.

Responsible editor: Niels Jungbluth

Electronic supplementary material The online version of this article (doi:10.1007/s11367-015-0961-0) contains supplementary material, which is available to authorized users.

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Keywords Dairy · Dietary guidelines · GHGE · LCA · Milk · Nutrition · Particulate matter · Sustainability

1 Introduction

Agricultural and food product systems present both an ideal and challenging application of life cycle assessment (LCA) methods due to their complex interlink between nature and technology. Significant progress has been made in the past decade in overcoming many of the challenges of food LCAs

(Roy, et al. 2009), and interest in their application has increased, as evidenced by the quality/quantity of work and growing attendance at the International Life Cycle Assessment of Foods Conference. LCA applied at the diet level is itself a burgeoning research area that is revealing much about the environmental impacts of dietary patterns (Heller et al. 2013; Hallström et al. 2015). The “use stage” nutritional effects of food—both impacts and benefits—have to date not been satisfactorily included in LCA.

Efforts to date to include nutritional aspects in food-related LCA have primarily focused on defining the functional unit (Heller et al. 2013). One review recommends a “quality-corrected functional unit” that takes the nutrient content of the food products into account (Schau and Fet 2008). Such an approach has become common place in farm-level LCAs of milk production when considering fat- and protein-corrected milk (International Dairy Federation 2010) and has been used to consider the effect of production practices on the protein content of wheat (Charles et al. 2006). Functional units based on a single nutritional aspect (e.g., protein content or caloric energy) are common and can be effective in particular inquiries (Gonzalez et al. 2011; Reijnders and Soret 2003). Still others have explored the use of nutritional profiling algorithms, which aggregate multiple nutritional dimensions into a single score, as the basis for functional unit (Smedman et al. 2010; Saarinen 2012; Heller and Keoleian 2012). Attempting to force impacts and benefits of nutrition into the functional unit can, however, create conceptual dissonance within an LCA framework built on expressing impacts in the numerator and positive outcomes (function) in the functional unit denominator.

Within nutritional sciences, studies typically focus on the health effects related to dietary intake and rarely account for environmental impacts occurring throughout the life cycle of food production. There is a growing appreciation, however, of the need to base dietary guidelines on nutritional as well as environmental science (van Dooren et al. 2014). Nutritional studies evaluate effects on various levels: individual nutrients (e.g., the effects of saturated fat, sodium, or dietary fiber), individual foods/food groups (e.g., the impacts of red meat or fruits and vegetables), and diets/dietary patterns (e.g., current diets vs. recommended diets). Nutritional epidemiology studies provide data that directly relate food items, nutrients, dietary patterns, or dietary quality indices to human health outcomes, typically expressed as disease incidence or mortality (Willett 2012). Such nutritional epidemiology-based information is captured by the Global Burden of Disease (GBD) study, a recent large-scale and detailed scientific effort to quantify global levels and trends in health.

The 2010 GBD reports detail the disease risk and impact in deaths and in Disability Adjusted Life Years (DALYs) associated with various risk factors (Murray et al. 2013a, b; Lim et al. 2012). The top ten leading risk factors globally that contribute to the GBD, in decreasing impact measured in

deaths, are dietary risks, high blood pressure, smoking, household air pollution, high body mass index, high fasting plasma glucose, ambient particulate matter (PM) pollution, physical inactivity, alcohol use, and high total cholesterol (IHME 2013). Food items and dietary habits are related to several of these risk factors either directly (dietary risks) or indirectly through agricultural production methods (ambient PM pollution). Interestingly, dietary risks are the largest contributor to both deaths and DALYs globally; in the US these contribute to 14 % of DALYs and 26 % of deaths (IHME 2013). Ambient PM pollution ranks seventh in terms of risk factors contributing to deaths globally and ninth in the US, leading to 2.2 % of total DALYs and 3.9 % of total deaths in the US (IHME 2013). A full list of global and US burden of disease for the top ten risk factors can be found in supporting information (SI), Section S1: Table S1.

In this paper, we first present a framework enabling a comprehensive assessment of both the direct nutritional and indirect environmental effects of food items/diet on a comparable scale. We then demonstrate the proposed framework with a fluid milk consumption case study, analyzing the potential effects of dietary substitution scenarios resulting from increased milk consumption. Focusing on human health as the “area of protection” for this case study, we utilize two relevant environmental impact categories on human health, global warming and particulate matter formation, and supplementing the 2010 GBD and other epidemiological data, we directly compare the environmental and nutritional impacts of a specific diet change on human health. Finally, we make recommendations for methodological and data developments necessary for a more complete comparison between environmental and nutritional effects of food production and consumption.

2 Methods

2.1 Combined nutritional and environmental LCA framework of diet

Figure 1 schematically outlines the proposed Combined Nutritional and Environmental LCA (CONE-LCA) framework, evolving out of conceptual outlines presented in Heller et al. (2013), for harmonizing nutritional and environmental effects over food life cycles. Food items, alone or as a part of a diet, are first associated with environmental emissions occurring over the life cycle of the food item (supply, production, and distribution stages), some of which may lead to population-scale health impacts. Likewise, consuming foods, the “use stage” of a food LCA, results in population-scale positive and/or negative nutritional health effects.

In this framework, the environmental assessment follows a traditional LCA approach. Starting from a common comparison basis (functional unit), emissions (e.g., N_2O , CH_4 , CO_2 ,

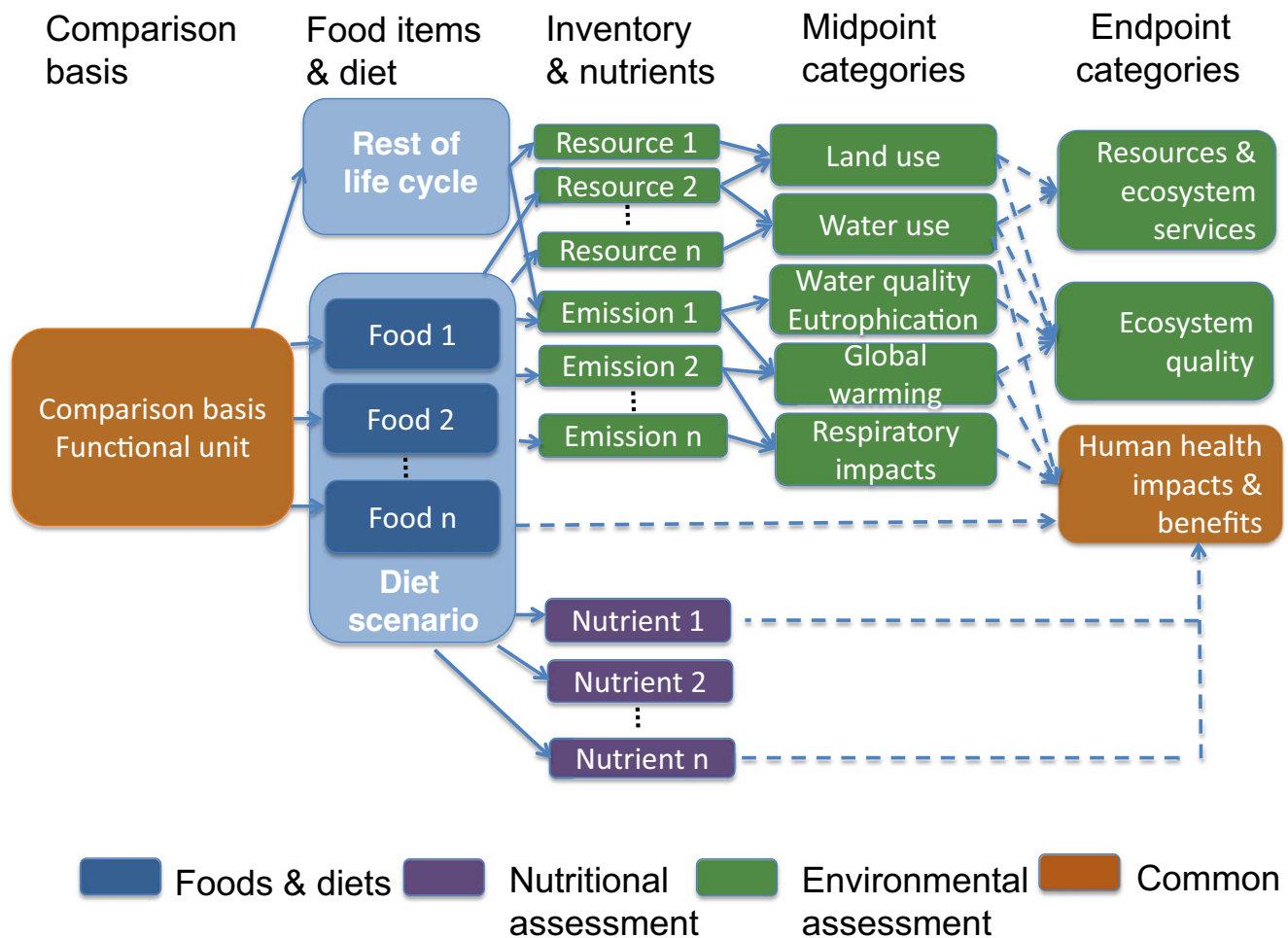


Fig. 1 Graphical representation of the combined nutritional and environmental health impact LCA framework. Dashed lines represent links between midpoint and endpoint categories that are useful to interpret impact scores, but whose quantification is also associated with a high degree of uncertainty

NH₃, PM_{2.5}) and important resource usages/extractions (e.g., water, land, mineral and energy) are determined, and then midpoint impacts in the most relevant categories are assessed: e.g., climate change, water consumption and quality (eutrophication), land use, and respiratory impacts. These impacts can then be linked to endpoint damages on human health, ecosystem quality, resource use, and ecosystem services (Jolliet et al. 2003; Jolliet et al. 2004). Human health damage, which is the focus of this paper, can ultimately be expressed as an impact in DALYs using epidemiological studies.

Nutritional impacts and benefits are assessed in parallel to environmental impacts relating the “use stage” in a food life cycle framework. Published epidemiology data are used to directly relate the food in question to reported nutritional health effects expressed in DALYs. When direct correlations are not available, the quantity of individual nutrients (e.g., protein, calcium, vitamin A, vitamin C, saturated fat, sodium) contained in foods can be calculated based on standard nutrient databases, and nutrients may then be associated using epidemiological data to overall health impacts or benefits.

Having both the environmental and nutritional assessment of food items or diets expressed in DALYs, as proposed by the CONE-LCA approach, allows the addition of nutritional assessment into a life cycle impact assessment framework for a parallel comparison of effects. However, it should be emphasized that the validity of results produced by the proposed approach is contingent on the data used, their availability, level of detail, and associated uncertainty.

2.2 Case study: milk consumption in the US

We tested the CONE-LCA framework by analyzing a case study of increased fluid milk consumption in the US as recommended by the *Dietary Guidelines for Americans 2010* (USDA and USHHS 2010). Since dietary recommendations are based on nutritional health benefits, it was interesting to explore in a parallel comparison the potential trade-offs between indirect environmental health impacts and nutritional health benefits associated with increased consumption of dairy.

We focused on *two environmental midpoint impact categories* of high environmental significance to the food production sector, global warming (GW) and respiratory inorganics. Global warming impacts from the dairy industry, indicated as greenhouse gas emissions (GHGE), are largely connected to methane (CH₄) and nitrous oxide (N₂O) emissions from enteric fermentation, manure management, and feed production (Thoma et al. 2013; Asselin-Balencon et al. 2013). Ammonia (NH₃) emissions from manure management contribute to the formation of secondary PM, adding to direct PM emissions from tractors and transportation (Henderson et al. 2013), that result in respiratory inorganic impacts. Impacts due to PM often dominate human health impacts in LCA, and PM is the most important environmental risk factor in the US, according to the GBD (Murray et al. 2013a). Henderson et al. (2013) found that other impacts from the milk life cycle on human health, such as toxicological effects and pesticide residues detected in milk, are limited compared to PM impacts. For the nutritional health impact assessment, we focused on both positive and negative health outcomes that have been associated with milk intake in epidemiological studies.

For the purposes of this case study, we use a serving of fluid milk as a functional unit. We also consider two alternative scenarios in which milk substitutes for other food items in the diet. The sections below define the dietary scenarios evaluated, describe the current average US diet baseline used in this case study, and summarize the environmental and nutritional assessment approaches used.

2.2.1 Dietary scenarios

The USDA-maintained 2010 Loss Adjusted Food Availability (LAFA) dataset (USDA ERS 2012) shows that the average adult US diet includes 148 g of fluid milk per day (0.61 servings) as part of the currently consumed 1.53 servings of dairy per person per day (USDA ERS 2012), which is about half the value recommended in the *Dietary Guidelines for Americans 2010*. For this study, we defined fluid milk as the consumption weighted average (with respect to population-scale consumption frequencies) of whole, 2 % reduced fat, 1 % reduced fat, and non-fat milk consumed in the US (detailed in the Electronic Supplementary Material, Section S2: Table S2).

We investigated the total effect resulting from a one-serving (244 g) increase in fluid milk consumption in the US. This addition led to a total daily consumption of 392 g, which is below the GBD-reported theoretical-minimum-risk exposure of 450 g/day for colorectal cancer (Lim et al. 2012). Adding one serving of fluid milk to the average adult US diet may or may not substitute a compensatory dietary portion of other food or beverages. To address the possibility of a potential substitution, we use a default iso-caloric equivalent basis for this case study as a first proxy and a pragmatic measure of

a substitution scenario. We acknowledge that a potential substitution scenario could be based on other rationale. Ideally, data from detailed market-based surveys should be used, when available, to identify and assess more realistic substitution scenarios. Section 4 further discusses the selection of substitution scenario and its consequences.

One serving of fluid milk has a nutritional energy content of 119 calories. Hence, we also investigated two replacement scenarios, assuming an iso-caloric substitution of 119 calories. Starting from the current average diet as a baseline, we investigated the following per person daily dietary scenarios (Fig. 2):

- A. Add a serving of fluid milk, with no change to the rest of the diet. This scenario results in an increased caloric intake over the average diet baseline.
- B. Add a serving of fluid milk while subtracting an equal caloric quantity from the overall average diet, excluding fluid milk. The resulting diet would be iso-caloric with the baseline average diet.
- C. Add a serving of fluid milk while subtracting an equal caloric quantity of sugar-sweetened beverages (SSB). The resulting diet would be iso-caloric with the baseline average diet.

2.2.2 Defining the current US diet

To understand how a shift in dairy consumption may affect the overall nutritional intake, we first characterized the current average US diet to establish a baseline for the environmental impacts. To define average US food consumption (average US diet), we used the 2010 LAFA data series, which tracks the availability of food commodities in the US marketplace (USDA ERS 2012). In the LAFA series, the available supplies for over 200 commodities are adjusted by percent loss assumptions at primary, retail, and consumer levels to arrive at a proxy for per capita food consumption in the US. The data are presented at the food commodity level (i.e., raw farm products like wheat and corn rather than consumables like bread or tortilla chips), which is far more manageable from an environmental impact perspective since most LCA studies of food are performed at this level. LAFA data also explicitly account for supply chain losses, allowing differentiation between foods produced, which contribute to environmental impact, and foods consumed, which are responsible for nutritional health effects.

2.2.3 Environmental health assessment

Greenhouse gas emissions GHGE associated with the average US diet were estimated via a meta-review of food LCA studies, according to Heller and Keoleian (2015). Several



Fig. 2 Graphical representation of dietary scenarios. The width of food groups corresponds to their caloric contribution to the total average US diet

limitations were identified in this approach: many underlying studies were not specific to US production scenarios, significant variability exists between studies of the same food item, transport and other variables were not treated in a consistent manner across studies. Still this approach captured the foods in the LAFA dataset, accounting for both consumption and losses, and utilized the most thorough collection of data currently available. For SSB, the GHGE estimate was based on an emission factor of 0.37 kg CO₂-eq/kg “soda” accounting for packaging and transportation to retail outlet (Vieux et al. 2012) and the corresponding energy content of 410 calories/kg “carbonated beverage” (USDA 2011). The fluid milk GHGE were obtained from a detailed LCA study specifically on US milk (Thoma et al. 2013). GHGE estimates were based on the IPCC Global Warming Potential for 100 years (IPCC 2007). Though estimates of human health impact of GW are much more uncertain than the midpoint indicator based on radiative forcing, an initial estimate of human health impact on a 100-year horizon is 0.82 μ DALY/kg CO₂-eq (Bulle et al., manuscript in preparation), which is useful for comparing on the same scale the order of magnitude of GW impacts with other environmentally induced effects on human health. The uncertainty associated with this impact factor is estimated at a GSD² of 4.8 with the uncertainty analysis description available in the Electronic Supplementary Material (Section S4).

Particulate matter Primary and secondary fine particulate matter, particles of less than 2.5 μ m diameter (PM_{2.5}), are among the environmental contributors to human health impacts of typical food and agricultural systems. In a comprehensive LCA of US fluid milk production, PM was found to

be the primary contributor to human health impacts, exceeding other environmental contributors considered by at least two orders of magnitude (Henderson et al. 2013).

PM_{2.5} and precursors (NO_x, SO₂, and NH₃) emissions are not routinely reported in food LCA studies, hindering direct estimates of emissions associated with the average US diet and SSB. To get around this limitation, we correlated PM-related emissions (primary PM_{2.5}, NO_x, SO₂, and NH₃) to the GW indicator measured in kg of CO₂-eq using 47 food-related Ecoinvent processes (Frischknecht et al. 2005). Results of the correlation analysis are summarized in Table 1 (detailed methodology description available in the Electronic Supplementary Material, Section S3). The correlations to CO₂-eq were found to be high for NO_x ($R^2=0.96$), primary PM_{2.5} ($R^2=0.92$), and SO₂ ($R^2=0.65$) and were thus used to estimate the corresponding emissions associated with the average diet and SSB (Table 1). For NH₃, we used other sources to estimate emissions since the correlation with CO₂-eq was weak ($R^2=-0.02$). The NH₃ emission estimate for the average diet was based on food-specific emission factors by Meier and Christen (2013). For SSB, emissions were estimated based on an emission factor for “sugar” from the same study as a proxy. All PM-related emissions for fluid milk were based on emission factors from the Comprehensive LCA of Fluid Milk (Henderson et al. 2013). The emission estimates for each 119-calorie equivalent portion intake are available in the Electronic Supplementary Material (Section S3: Table S4).

Overall PM_{2.5} emissions in the US are responsible for 103,000 deaths per year and 1,820,410 DALYs, mostly via cardio-pulmonary diseases (Murray et al. 2013a). To link

Table 1 Summary of results from correlation of PM-related emissions to GHGE for all food-related processes in the Ecoinvent database and characterization factors for PM species

	PM _{2.5}	SO ₂	NO _x	NH ₃
Emission correlation analysis model: $y=b \times x^a$				
CO ₂ -eq correlation factor				
(kg/kg CO ₂ -eq)	2.4E-4	8.3E-4	2.7E-3	3.5E-3 ^b
GSD ²	1.5	2.9	1.5	6.8
R ²	0.92	0.65	0.96	−0.02
Human Health Impacts per kg emitted				
Characterization factor				
(kgPM _{2.5} -eq/kg emitted)	1.2E-3	5.2E-2	1.1E-2	1.1E-1
(DALYs/kg emitted)	3.0E-4	6.2E-5	1.3E-5	1.3E-4

^a y =precursor, x =CO₂-eq, b =CO₂-eq correlation factor

^b Not used in this analysis

PM_{2.5} emissions to human health impacts, we used the framework described by Fantke et al. (2014). A set of default human intake fractions (iF) associated with PM-related emissions (Humbert et al. 2011) was complemented with corresponding effect factors to yield characterization factors (CFs), i.e., human health impacts per kilogram of PM-related emission, accounting for cardiopulmonary and lung cancer impacts from both primary and secondary PM_{2.5} (Gronlund et al. 2015). Using such epidemiological data to link midpoint (PM_{2.5} formation) to endpoint impact categories (human health impacts) supports comparability between environmental and nutritional findings in our study. The CF used in calculating health impacts for primary PM_{2.5} is 3.0×10^{-4} DALYs/kg PM_{2.5} for emissions in rural areas; likewise, for secondary PM_{2.5}, the CF estimates for SO₂, NO_x, and NH₃ are 6.2×10^{-5} DALYs/kg SO₂, 1.3×10^{-5} DALYs/kg NO_x, and 1.3×10^{-4} DALYs/kg NH₃, respectively (Table 1). Uncertainty factors for the CFs as well as all other estimates used for the PM health impact calculation are available in the Electronic Supplementary Material (Section S4: Table S7).

2.2.4 Nutritional health assessment

Fluid milk There is a number of positive and negative health outcomes linked with milk consumption. One benefit of milk consumption on human health is related to a reduced risk of colorectal cancer. Another effect is related to stroke. Limited evidence also suggests that milk intake is associated with an increased risk of prostate cancer in males. Increased milk consumption has also been associated with a change in body mass index (BMI). However, evidence has been inconclusive with two recent meta-analyses of epidemiological studies and randomized clinical trials indicating that increased dairy consumption does not have an effect on weight (Abargouei et al. 2012) or has a modest effect on weight loss (Dougkas

et al. 2011). Hence, we have decided to not include BMI change as a health outcome in our nutritional assessment. A further discussion of milk consumption-related outcomes is provided in Section 4.

Colorectal cancer: Milk intake has been found to reduce the risk of colorectal cancer as concluded by the World Cancer Research Fund/American Institute for Cancer Research (WCR/AICR 2007). The hypothesized mechanism for this inverse association is related to the high calcium content of milk through two possible pathways. The first is the reduced cell proliferation in the colonic epithelium when calcium is bound to pro-inflammatory secondary bile and ionized fatty acids. The second is the enhanced cell differentiation between normal and apoptotic cells under the influence of calcium in a number of intracellular influences (Aune et al. 2012). This inverse association between milk intake and colorectal cancer is supported by multiple meta-analyses that report a statistically significant association in a consistent manner and with almost equal risk ratio (RR) estimates (Murray et al. 2013a; Aune et al. 2012). To assess the corresponding nutritional benefit, measured in avoided DALYs related to colorectal cancer, attributable to an increase of current fluid milk consumption by one serving, we used a 1.11 RR (95 % CI 1.03–1.20) per 226.8 g/day of milk intake decrease reported in the 2010 GBD (Murray et al. 2013a), a theoretical-minimum-risk exposure of 450 g/day (Lim et al. 2012), above which it is assumed that there is no additional health benefit, and a US colorectal cancer burden estimate of 1,146,830 DALYs for both sexes in 2010 (IHME 2013).

All stroke outcomes: Evidence from a number of epidemiological studies suggests a protective effect of milk intake to all stroke outcomes (Elwood et al. 2004; Larsson et al. 2012; Ness et al. 2001). The hypothesized mechanisms of this association are linked to the influence of milk intake on hypotension (Larsson et al. 2012) and blood cholesterol levels (Elwood et al. 2004). A more recent meta-analysis of total milk intake and any stroke outcome generated a statistically significant summary association of 0.85 RR (95 % CI 0.77–0.94) comparing high (two to four servings) versus low (0–0.5 servings) milk intake (Alexander et al., manuscript submitted for publication), approximately corresponding to an intake difference of 541 g of milk (95 % CI 400–732). We used this RR to estimate avoided DALYs related to stroke when current fluid milk consumption is increased by one serving, assuming there is no additional benefit above an intake of 597 g, and a 2010 US stroke burden of disease estimate of 1,569,720 DALYs for both sexes (IHME 2013).

Prostate cancer: Limited evidence suggests that there could be an increased risk of prostate cancer in males

associated with milk consumption, with the most cited hypothesized mechanism being that calcium may increase risk by disrupting the circulation of vitamin D in the human body (WCR/AICR 2007; Aune et al. 2015). Although conclusions are less established and confounded by conflicting evidence for this health outcome, a recent meta-analysis reported a summary RR for a 200-g/day increase of milk intake of 1.03 (95 % CI 1.00, 1.06) with some indication of a nonlinear association; risk increased rapidly from 0 to 100–200 g milk/day with no further risk increase above this level (Aune et al. 2015). Hence, given a current milk intake of 148 g/day, adding one serving of fluid milk to the average US diet may not result in an increase in the burden of prostate cancer according to these findings. However, taking a conservative approach, and allowing for an example of a possible negative nutritional effect of a food/diet, we considered this health endpoint in our nutritional health assessment using the RR by Aune et al. (2015) and a 2010 US prostate cancer burden of disease of 592,400 DALYs (IHME 2013).

Average diet The average diet, as defined in our study, is consisted of numerous food items that might have positive or negative effects on human health such as fruits and vegetables and SSB, respectively. For the purposes of this case study, we assume that average diet has no effect on health and our results and conclusions are contingent on this assumption.

Sugar-sweetened beverages One important dietary risk factor for the US is “diet high in sugar-sweetened beverages”. The US burden of disease for this risk factor includes health outcomes such as diabetes (75 %), cardiovascular and circulatory diseases (20 %), and cancer and musculoskeletal disorders (3 % each), as reported in the 2010 GBD (IHME 2013). According to Lim et al. (2012), the theoretical-minimum-risk exposure for SSB is 0 g/day. Therefore, the total burden of disease related to SSB of 770,584 DALYs reported in the 2010 GBD (IHME 2013) can be considered to be the direct outcome of the current US SSB consumption, estimated at 236 calories/person/day (Han and Powell 2013). Building on these findings, we estimated an overall SSB-related disease effect factor of 0.03 μ DALYs/SSB-calorie (95 % CI 0.02–0.04).

3 Results

3.1 Environmental health assessment

3.1.1 Greenhouse gases

The GHGE associated with the current average diet of an American consumer (assuming an intake of 2534 calories/

day according to LAFA data) was 5.0 kg CO₂-eq/person/day (95 % CI 2.5–9.2). To demonstrate the impact associated with our three diet modification scenarios, the bars in Fig. 3 present the GW midpoint impacts associated with a 119-calorie equivalent portion of three distinct components: fluid milk, average diet, and SSB.

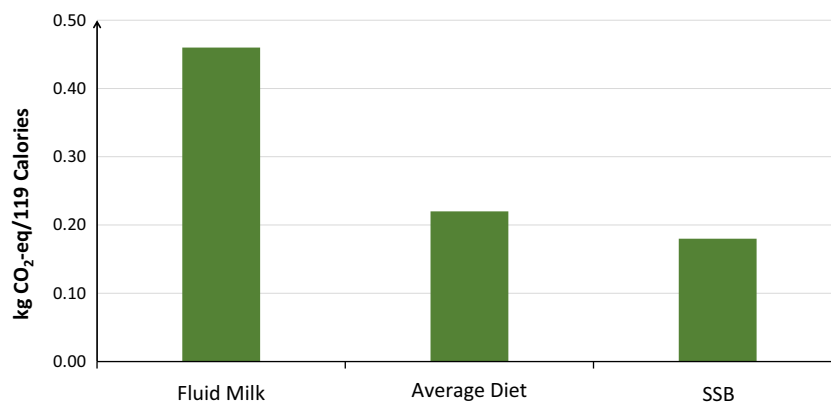
Fluid milk produced the highest GW midpoint impact per calorie; a serving of fluid milk (scenario A) was associated with 0.47 kg CO₂-eq. The corresponding human health endpoint impact was estimated at 0.38 μ DALY (GSD²=4.9). In comparison, the GW impacts of 119 calories of average diet amount to 0.24 kg CO₂-eq which was equivalent to 0.19 avoided μ DALY (GSD²=5.5), close to half the impact of fluid milk. Therefore, the net GW impact associated with scenario B could be calculated as the difference between the impact of one serving of milk and the impact of the iso-caloric substitution of average diet, i.e., 0.47–0.23=0.24 kg CO₂-eq (0.19 μ DALY). Finally, 119 calories of SSB were linked to 0.19 kg CO₂-eq, equal to 0.15 avoided μ DALY (GSD²=6.5). As a result, the net difference in GW impact between milk and SSB for scenario C was 0.29 kg CO₂-eq (0.23 μ DALY).

3.1.2 Particulate matter

The total respiratory inorganic impacts from PM-related emissions linked to the baseline US diet were 2.2 g PM_{2.5}-eq/person/day (95 % CI 1.1–3.9). Figure 4 illustrates the PM-related emissions in grams (g) PM_{2.5}-eq corresponding to 119 calories of fluid milk, average diet, and SSB. The emissions for scenario B and C can be estimated by subtracting the emissions associated with each corresponding substitution from those of fluid milk.

One serving of fluid milk (scenario A) was linked to total PM-related emissions of 0.26 g PM_{2.5}-eq, corresponding to a human health impact of 0.32 μ DALY (95 % CI 0.04–0.83). This impact was predominately caused by NH₃ emissions from barn and manure management (80 %) and to a lesser extent by SO₂ (13 %). The iso-caloric equivalent of the average diet resulted in emissions about half of those of fluid milk. Thus, 70 % of the PM-related impact for the average diet was due to NH₃, 14 % was due to primary PM_{2.5}, and 10 % was due to SO₂. The corresponding human health impact was 0.15 avoided μ DALY (95 % CI 0.02–0.39). This resulted in a net PM emissions estimate of 0.13 g PM_{2.5}-eq, equal to 0.15 μ DALY for scenario B. For the 119-calorie SSB equivalent, we estimated PM-related emissions of 0.03 g PM_{2.5}-eq, one third of which was attributable to primary PM_{2.5}, which was analogous to a human health damage of 0.04 avoided μ DALY (95 % CI 0.01–0.10). As a result, the net PM-related emissions estimate for scenario C was 0.23 g PM_{2.5}-eq, corresponding to a health impact of 0.28 μ DALY, slightly lower than scenario A.

Fig. 3 Global warming impacts measured in kg CO₂-eq associated with a 119-calorie equivalent portion of three distinct intakes: (1) fluid milk, (2) average diet, (3) sugar-sweetened beverages



3.2 Nutritional health assessment

We estimated a linear dose–response relationship between milk intake in g/person/day and the impact of colorectal cancer and stroke in DALYs/person/day (Fig. 5a, b). For prostate cancer, the dose–response for the male population is displayed in Aune et al. (2015).

We used these dose–response functions to estimate the nutritional human health burden change for each outcome as a result of a shift in fluid milk intake of one additional serving to the current consumption. For colorectal cancer, we found that the addition of one serving of fluid milk results in a benefit of 1.10 μ DALY (95 % CI 0.78–1.56), while for all stroke outcomes the benefit was equal to 0.95 μ DALY (95 % CI 0.67–1.35). In parallel, we estimated an increase in prostate cancer burden for the male population equal to 0.32 μ DALY (95 % CI 0.20–0.51). To extrapolate the prostate cancer impact to the overall population, we accounted for the fraction of males in the US; this resulted in an impact of 0.16 μ DALY (95 % CI 0.10–0.26). Overall, we estimated that there is a net nutritional benefit of 1.88 μ DALY (95 % CI 0.03–1.02) in response to

the addition of one serving of fluid milk to the current average US diet.

The nutritional benefit from dietary changes could be further increased when food items with negative health effects such as SSB are replaced. In particular, a reduction of 119-calorie equivalent in the SSB daily intake was associated with 3.48 avoided μ DALY (95 % CI 2.23–5.43). This finding demonstrates the importance of considering the added nutritional effects (positive and/or negative) of substitution scenarios when dietary changes are evaluated.

3.3 Overall comparison

Figure 6 represents the overall comparison of environmental and nutritional effects for the three scenarios, all expressed at the human health endpoint level in avoided μ DALYs/person/day (result summary available in the Electronic Supplementary Material, Section S5: Tables S9 and S10). Based on the epidemiological evidence considered, we estimated that adding one serving of fluid milk to the present adult average U.S. diet leads to an overall health benefit for scenario

Fig. 4 Particulate matter related emissions measured in grams (g) PM_{2.5}-eq associated with a 119-calorie equivalent portion of three distinct intakes: (1) fluid milk, (2) average diet, (3) sugar-sweetened beverages

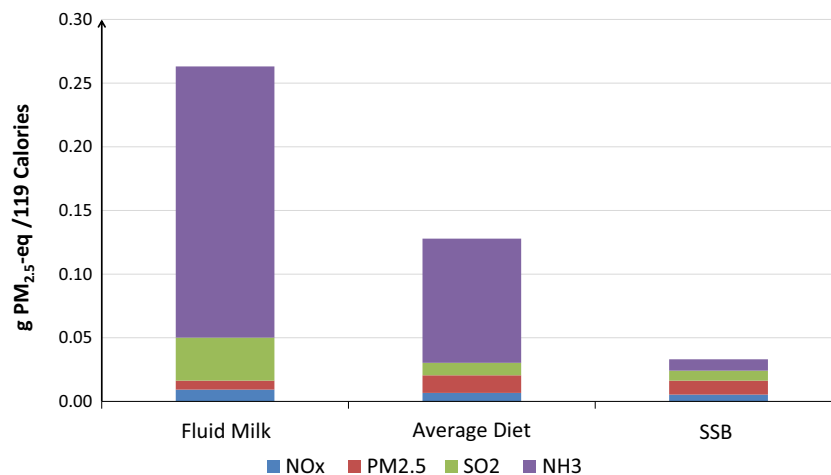
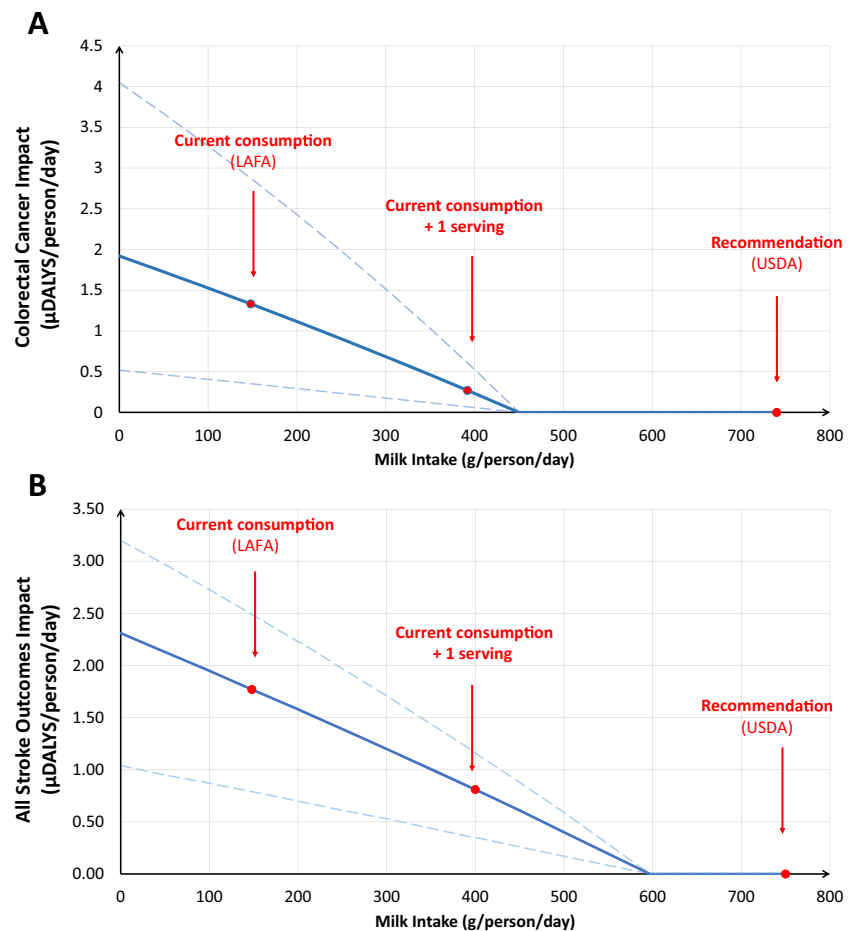


Fig. 5 Dose–impact function for milk intake and **a** colorectal cancer impact and **b** stroke, with 95 % confidence intervals shown as dashed lines



A. Considering no nutritional health effect associated with the average diet substitution in scenario B, this overall health

benefit was slightly increased. This benefit was substantially increased when considering the avoided impact associated

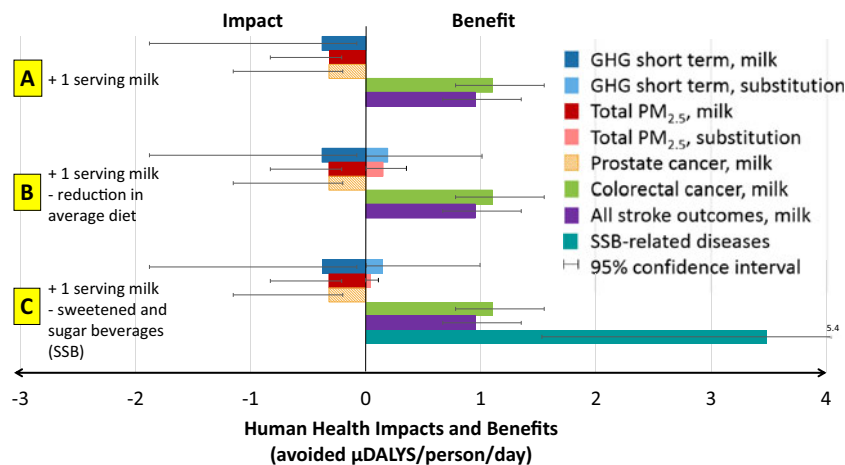


Fig. 6 Comparison of daily environmental impacts (GHGE and PM-related emissions) and nutritional effects (milk-related disease: colorectal cancer, stroke, prostate cancer for males only, and SSB-related diseases) in avoided μ DALY/person/day, for three dietary changes scenarios over the average diet: (1) add one serving of fluid milk (scenario A), (2) add one serving of fluid milk with an iso-caloric substitution from the overall average diet (scenario B), and (3) add a serving of fluid milk with an iso-caloric substitution from sugar-

sweetened beverages (scenario C). A positive value indicates a benefit (avoided burden), while a negative value indicates an impact (induced burden). Error bars indicate the 95 % confidence intervals representing a preliminary characterization of uncertainty. It should be noted that for scenario B, we assumed that substitution from the average diet has no nutritional health effects, with all observed nutritional health effects solely associated with the addition of one serving of milk

with the SSB substitution in scenario C. To estimate the robustness of this finding, we employed the approach by Hong et al. (2010) for each scenario to estimate the likelihood that overall benefits are greater than overall impacts, accounting for uncertainty propagation. We estimated that there was at least a 98.1 % probability that overall benefits of one additional serving of fluid milk exceeded the corresponding impacts in scenario A. This probability increased to 99.2 % for scenario B and 100 % for scenario C. When considering only outcomes reported in the 2010 GBD for the nutritional assessment, colorectal cancer for milk intake and all diseases related to SSB intake, we observed greater likelihood variability between scenarios. In fact, the increased consumption of fluid milk remained beneficial under all scenarios with a probability of 80.8 % for scenario A, 91.4 % for scenario B, and 100 % for scenario C. As a sensitivity analysis, we restricted the analysis to the male US population, assuming that all impact categories besides prostate cancer have an equal effect on males and females. This restriction resulted in a slightly reduced probability variability of 98.9 and 99.7 % for scenarios A and B, respectively, while the probability remained 100 % for scenario C.

4 Discussion

In this paper, we describe the CONE-LCA framework that allows for a parallel epidemiology-based assessment in LCA of environmental and nutritional effects of food items. We demonstrate this approach using a proof-of-concept case study investigating the environmental and nutritional effects associated with an increase in US milk consumption. Our analysis was only focused on human health as the “area of protection”, using DALYs as a common endpoint metric, since nutritional effects only contribute to this area via the “use stage” of a food LCA. We limited our analysis to only two relevant environmental impact categories contributing to human health, namely, PM and GW impacts. However, it should be emphasized that the CONE-LCA framework can be extended to other human health-related environmental impact categories so as to provide a more balanced and comprehensive assessment.

Specific to this case study, nutritional human health benefits associated with the addition of one serving of fluid milk exceeded the corresponding overall impacts (environmental and nutritional) under three dietary scenarios. It should be emphasized that our findings are initial estimates and dependent on the present quality and high uncertainty level of the available data. Hence, our findings should be interpreted with caution and only within the context of this study. This is reflected in the high level of uncertainty for the different assessment categories (Electronic Supplementary Material, Section S4: Tables S5–8). A detailed Monte Carlo analysis of uncertainties and trade-offs would be useful in further refining these findings, as well as obtaining more accurate

estimates of the uncertainty ranges for the considered impacts and benefits.

We found that one serving of fluid milk has a higher impact in both environmental categories under consideration, GW on a 100-year time horizon and respiratory inorganics formation, than the iso-caloric equivalent from the average diet and SSB. This resulted in lower overall environmental impacts at the human health endpoint in DALYs associated with increased fluid milk consumption when substitutions are considered (scenario B and C), compared to a no-substitution case (scenario A).

The characterization factors for human health impacts due to GW are very uncertain. Hence, the corresponding human health impact estimates should only be considered as order of magnitude approximations. In future work, it would be informative to also assess the “long-term” GW human health impacts beyond the 100-year time horizon.

Estimates of respiratory inorganics have wide confidence intervals, mostly due to the uncertainty on the CFs originating from variability in iF estimates. Additionally, there is a potential for double-counting the impact on human health due to the way secondary $PM_{2.5}$ is currently handled in LCA. NO_x , SO_2 , and NH_3 impacts are estimated using individual characterization factors that do not consider the interactions between these PM precursors. A better characterization of secondary $PM_{2.5}$ in LCA is therefore needed, especially for food items with high NH_3 emissions. This may be achieved through a spatially differentiated analysis of the respective effects of NO_x , SO_2 , and NH_3 on the formation of secondary $PM_{2.5}$ which depend on the each precursor’s background concentrations and the location of emission, rural or urban, for which limiting factors for PM formation may differ.

Our nutritional analysis focused on health outcomes for which epidemiological evidence supports an association with milk and SSB intake. For the nutritional assessment of milk intake, the analysis was limited to only two beneficial (colorectal cancer and stroke) and one detrimental (prostate cancer) health outcome associations. It should be noted that evidence supporting an association between milk intake and increased prostate cancer risk in males is controversial. In the study by Aune et al. (2015), whole milk appears to have a protective effect on prostate cancer, whereas in the same study any milk and low-fat milk increase risk. The same study indicates a threshold effect where prostate cancer risk is flattened for intakes over 200 g/day of milk. Taking this into account, the prostate cancer burden for the male population corresponding to the addition of one serving of fluid milk over the present consumption of 146 g would be increased by only 0.08 μ DALY (95 % CI 0.05–0.13). However, for this case study, we included the worst case scenario to avoid any overestimation of the net nutritional benefit of milk. Regarding the framework proposed in this paper, it should be emphasized that the nutritional analysis of this case study demonstrates that the

CONE_LCA can evaluate foods that have both positive and negative effects on health.

In a further investigation of the nutritional health effects of milk, other health outcomes associated with milk and/or dairy product consumption should also be considered. For example, toxicological impacts due to pesticide residues detected in milk could be accounted for. However, based on data collected in 2005 from the USDA Pesticide Data Program, the estimated impact from intake concentrations of pesticides in one serving of milk is 0.00015 μ DALY (Henderson et al. 2013), dominated by the impact from dieldrin. This appears to be a negligible contribution compared to other “use stage” consumption impacts. Change in BMI is another health outcome to be considered with increased milk consumption without any compensatory decrease in caloric intake (scenario A) or increase in physical activity. Although the relationship between dairy consumption and weight loss/gain has been extensively studied, findings are inconclusive. Results from two recent meta-analyses show that increased dairy consumption does not have an effect on weight (Abargouei et al. 2012) or has a modest effect on weight loss (Dougkas et al. 2011). Other health outcomes that could be considered are cardiovascular diseases, type 2 diabetes, hypertension, osteoporosis, and breast cancer. Work by Alexander et al. (manuscript submitted for publication) provide RRs for some of these outcomes that suggest significant protective effects from total dairy intake that show more certainty than those of fluid milk alone. It would therefore be of interest to further investigate the environmental and nutritional health impacts and benefits from changes in total dairy consumption versus milk alone.

When assessing dietary changes, it is important to consider potential substitutions, especially when assessing nutritional effects, since substitutions may have positive or negative health effects. In our case study, we investigated increased milk consumption but also considered scenarios that assumed possible iso-caloric substitutions from the average diet and SSB. Beyond the exact choices of the substitution scenario (as many could be considered), our results primarily emphasize the importance of considering substitution scenarios when assessing dietary change. This is apparent in scenario C where the avoided nutritional health impact associated with SSB dominate the overall assessment (Fig. 6). It should be noted that although potentially significant, a nutritional assessment of the substituted average diet has not been performed. In this case study, average diet is defined using the 2010 LAFA dataset to primarily determine its environmental impacts of global warming and particulate matter. The composition consisted of various items that are also linked to various health outcomes with positive and negative effects. Although a single impact score of the average diet substitution could be theoretically possible by combining the effect of each single food items (consumption in average diet compared with recommended or reference values on epidemiological studies),

this was beyond the scope of this study since it could potentially be problematic due to double-counting resulting from the assumption of independent effects between food items and due to the limited data available. Instead, the average diet substitution was considered to have neither nutritional benefits nor impacts on human health. In a worst case scenario analysis where milk would substitute foods that are beneficial for human health, such as fresh fruits and vegetables, the overall assessment may lead to an overall health impact. Finally, we also acknowledge that the definition and iso-caloric parameterization of the substitution scenarios was pragmatic in this proof-of-concept exercise. The scenarios evaluated here are not exhaustive, and alternative approaches for selecting scenarios may be used. Ideally, detailed market-based data that capture consumer behavioral patterns should be used to determine potential substitutions resulting from dietary changes.

Finally, in this study, we have not considered the possible variations in health effects of increased milk consumption by sex or age. The health effects of milk likely differ between men and women either by magnitude or outcome, as evident by the nutritional assessment of prostate cancer. It is also possible that health effects might be different between age groups (younger adults versus the elderly). In this paper, we considered the general order of magnitude of potential effects at the population-scale level for an adult American with average dietary consumption. Therefore, a possible improvement of our analysis would be the segmentation by sex and age groups to allow for more refined environmental and nutritional human health effect estimates associated with increased milk consumption.

5 Conclusions and recommendations

The CONE-LCA framework proposed in this paper provides the groundwork for an improved, more balanced LCA methodology that, in addition to the production-related environmental impacts of food items, also considers for the “use stage” nutritional effects resulting from consumption. The originality of this framework is that it innovatively takes dietary epidemiological evidence expressed in DALYs and compares them consistently with life cycle impact assessment human health endpoint DALYs, related to food items. Assessing environmental and nutritional human health effects of food items/diet at the DALYs level aligns well with the traditional LCA approach as well as methods used for assessing burden of disease. The need and importance of such a framework was demonstrated with a proof-of-concept case study in which we compared the environmental and nutritional effects on human health associated with the daily addition of one serving of fluid milk to the current average US diet. Therefore, the approach proposed in this paper allows for a quantitative estimate of human health impacts by evaluating, via a common

metric, the complements and trade-offs between nutritional and environmental effects of foods/diets.

However, there is a need to maintain full transparency in communicating results generated with this framework due to possible limitations emerging from the quality of data used and more specifically the uncertainties associated with the characterization of environmental and nutritional effects of food items. Where the environmental assessment is concerned, it is important to differentiate the influence of each midpoint category on the endpoint and the related uncertainty. At the same time, epidemiological evidence for both environmental and nutritional effects should be used with caution as they are suggestive of correlations but do not imply causation. More specifically, it is possible that the magnitude of the effect associated to an individual risk factor is partially due to confounders, which at that point of the study were either unknown or not controlled for. For example, when estimating the effect of PM on human health, it is possible that noise also contributes to that effect since high PM levels due to traffic may be correlated to high noise exposures. The same applies to nutritional epidemiology; for instance, the burden associated with SSB might be confounded by sedentary lifestyle and/or higher consumption of less healthy food such as saturated fats, sugars, and sodium. All in all, there is a need to acknowledge uncertainties, knowledge gaps, and limitations when attempting to quantify the environmental and nutritional linkage to human health.

Despite the limitations and the exploratory character of this CONE-LCA framework proof-of-concept case study, several recommendations can already be drawn. First, our results demonstrate the potential limitations when using the established environmentally focused LCA approaches for food items and suggest that the nutritional effects of the “use stage” should be considered in food-related LCAs. In our case study, we have shown that the nutritional effects during the “use stage” could have an effect of comparable magnitude when compared to environmental effects. Second, the study emphasized the importance and need for enhanced data availability and refinement to support this new approach. Third, when considering the human health impacts of GHGE and PM as well as nutritional effects, results within the context of this study are suggestive of a health benefit linked with a one-serving increase in milk consumption for American adults, provided that it does not substitute any health-beneficial food items such as fruits and vegetables. The benefit is further increased when considering an iso-caloric substitution of less healthy beverage options. Further studies are now needed to test whether this conclusion holds within a more comprehensive assessment of environmental and nutritional health impacts, e.g., examining additional substitutions, such as the substitution of health-beneficial food items like fresh fruits and vegetables. Fourth, while not being the purpose of this paper, our results suggest that despite the focus on

GHGE of many food LCA studies other environmental health impacts of food items, such as PM, can also be relevant. Finally, we recommend applying this CONE-LCA approach, which constitutes the infancy of a new area within food/diet-related LCA, to other food items or diets to characterize potential trade-offs between environmental and nutritional impacts when making recommendations about sustainable diets and food choices.

Acknowledgments This work was funded by an unrestricted grant of the Dairy Research Institute (DRI), part of Dairy Management Inc. (DMI). We thank Dr. Dominik Alexander for his support in interpreting the nutritional meta-analysis.

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